🧠 DISK SCHEDULING (Complete Deep Notes)

🔷 1. INTRODUCTION

Disk scheduling OS ka ek important part hai jo decide karta hai ki **disk I/O requests ko kis order me serve kiya jaye**.

👉 Kyunki multiple processes disk access request karte hain, OS ko optimize karna padta hai.

💡 TECHNICAL DEFINITION:

Disk scheduling is the process used by the operating system to determine the order in which disk I/O requests are processed to minimize seek time and improve overall system performance.

🔷 2. WHY DISK SCHEDULING IS NEEDED?

- Multiple processes request disk access
- Disk head movement slow hota hai
- Seek time high hota hai
- Efficiency improve karna hota hai

🔷 3. IMPORTANT TERMS

📌 Seek Time

Time taken by disk head to reach track

📌 Rotational Latency

Disk rotation ka wait time

📌 Transfer Time

Data transfer time

📌 Disk Arm Movement

Head movement across tracks

🔷 4. GOAL OF DISK SCHEDULING

✓ Minimize seek time
✓ Increase throughput

✓ Reduce response time
✓ Fair allocation

🔥 5. DISK SCHEDULING ALGORITHMS (VERY IMPORTANT)

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1. FCFS (First Come First Serve)

🔷 **Concept:**

Requests ko order me serve karta hai jaisa aata hai

✓ **Example:**

Request queue: 98, 183, 37, 122, 14

✓ **Advantages:**

- Simple
- Fair

❌ **Disadvantages:**

- High seek time
- Slow performance

💡 **Interview:**

FCFS is a simple disk scheduling algorithm that serves requests in order of arrival without optimization.

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2. SSTF (Shortest Seek Time First)

🔷 **Concept:**

Jo request closest hai current head position se, wo serve hota hai

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✓ **Advantages:**

- Low seek time

- Better performance than FCFS

❌ **Disadvantages:**

- Starvation possible
- Far requests wait karte rehte hain

💡 **Interview:**

SSTF selects the disk request that requires the least seek time from the current head position.

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3. SCAN (Elevator Algorithm)

🔷 **Concept:**

Disk head ek direction me move karta hai aur sab requests serve karta hai, then reverse karta hai

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✔ **Example:**

Like elevator going up and down

✔ **Advantages:**

- No starvation
- Better performance

❌ **Disadvantages:**

- Middle requests advantage me rehte hain

💡 **Interview:**

SCAN algorithm moves the disk arm in one direction servicing requests until it reaches the end, then reverses direction.

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4. C-SCAN (Circular SCAN)

🔷 **Concept:**

Head only one direction me move karta hai, end pe jump karke start se restart karta hai

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✔ **Advantages:**

- Uniform wait time
- Fair distribution

❌ **Disadvantages:**

- Extra movement at reset

💡 **Interview:**

C-SCAN treats the disk as a circular list and services requests in one direction only, improving uniform response time.

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5. LOOK Algorithm

🔷 **Concept:**

SCAN jaisa hai but disk ke end tak nahi jata, last request tak jata hai

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✔ **Advantage:**

- Less unnecessary movement

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6. C-LOOK Algorithm

🔷 **Concept:**

C-SCAN ka optimized version, sirf last request tak jata hai then jumps

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✔ **Advantage:**

- Efficient than C-SCAN

🔥 **6. COMPARISON TABLE**

Algorithm	Seek Time	Starvation	Fairness
FCFS	High	No	High
SSTF	Low	Yes	Medium
SCAN	Medium	No	High
C-SCAN	Medium	No	Very High
LOOK	Low	No	High
C-LOOK	Lowest	No	Very High

🔷 7. ADVANTAGES OF DISK SCHEDULING

✓ Reduces seek time
✓ Improves performance
✓ Efficient disk utilization
✓ Faster response

🔷 8. DISADVANTAGES

❌ Complexity increases
❌ Some algorithms cause starvation
❌ Additional overhead

🔥 9. IMPORTANT IMPROVEMENTS / + POINTS

✓ SSTF is better than FCFS
✓ SCAN avoids starvation
✓ C-SCAN gives uniform wait time
✓ LOOK avoids unnecessary disk movement
✓ C-LOOK is most optimized

🧠 10. REAL LIFE ANALOGY

- FCFS → queue at bank
- SSTF → nearest counter first
- SCAN → elevator
- C-SCAN → circular elevator loop

🎯 11. IMPORTANT INTERVIEW QUESTIONS

Basic:

- What is disk scheduling?
- Why is it needed?

Algorithms:

- Explain FCFS, SSTF, SCAN
- Difference between SCAN and C-SCAN
- Difference between LOOK and C-LOOK

Advanced:

- Which algorithm is best and why?
- What is starvation in SSTF?
- Why C-SCAN is better than SCAN?